

Hospital Flow Burden in MIMIC-IV

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What this project does

This project uses hospital transfer and discharge records from **MIMIC-IV v3.1** to examine how long patients remain in individual care units at Beth Israel Deaconess Medical Center. The goal is to identify where unusually long care-unit segments consume capacity and where an operations team should investigate first.

It answers three practical questions:

1. **How much of the hospital's recorded care-unit time is concentrated in long segments?**
2. **Which care units contribute the most time beyond a 72-hour threshold?**
3. **Do discharge destinations provide a signal that downstream-care access may be associated with longer final segments?**

Executive finding

The burden is concentrated in the long tail. **14.1%** of valid system-wide care-unit segments exceed 72 hours, but those segments account for **65.3%** of all recorded segment-hours. The largest contributor to excess time beyond 72 hours is **Medicine**. The median final segment in Medicine is **57.6 hours longer** for institutional post-acute destinations than for home discharge.

System-wide median segment

8.8 h

Segments >48 h

20.5%

Segments >72 h

14.1%

Segments >7 d

4.0%

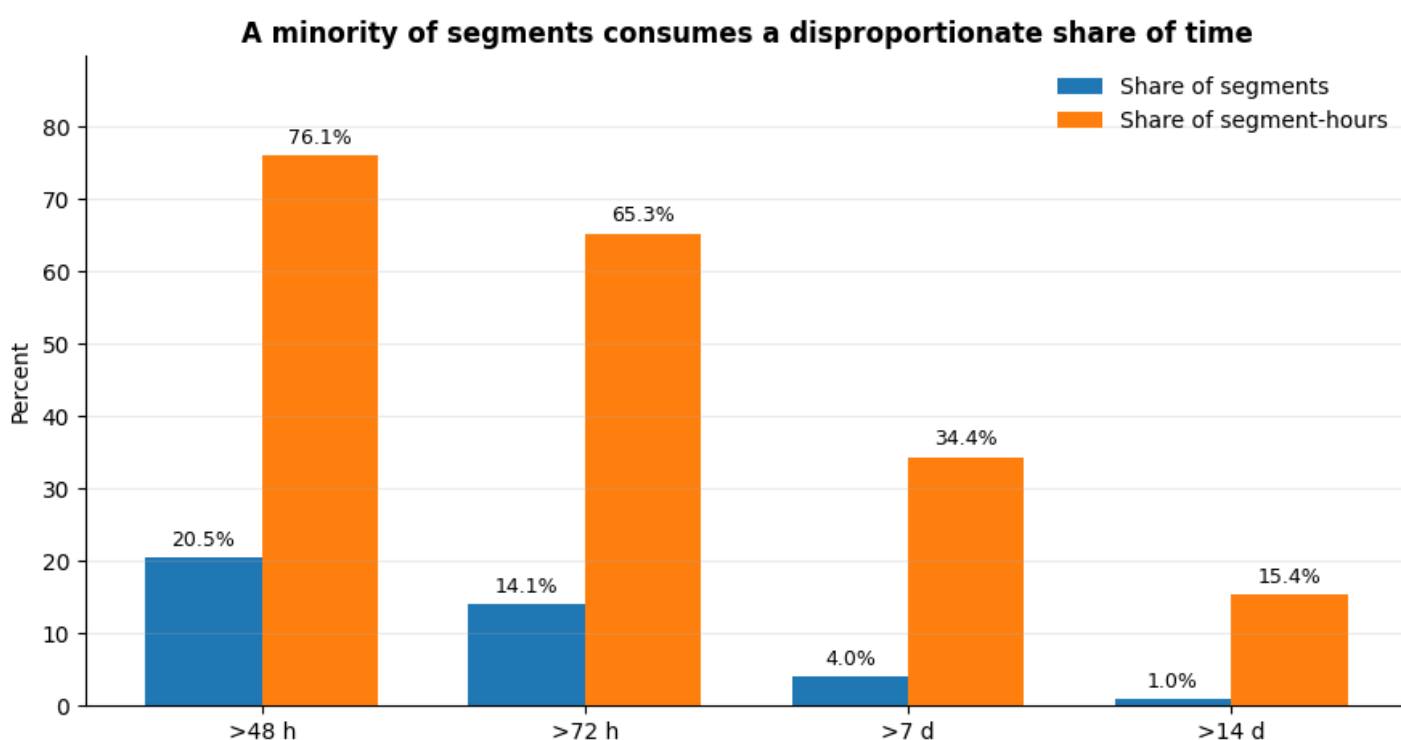
The difference between the median and the long-stay percentages is the central operational story: most segments are relatively short, but a meaningful minority remain for days.

1. Long segments consume a disproportionate share of capacity

The first graph compares two quantities at each duration threshold:

- **Share of segments:** the percentage of care-unit segments lasting longer than the threshold.
- **Share of segment-hours:** the percentage of all recorded care-unit time consumed by those segments.

When the segment-hours bar is much taller than the segment-share bar, a small group of long segments is consuming a disproportionate amount of capacity.

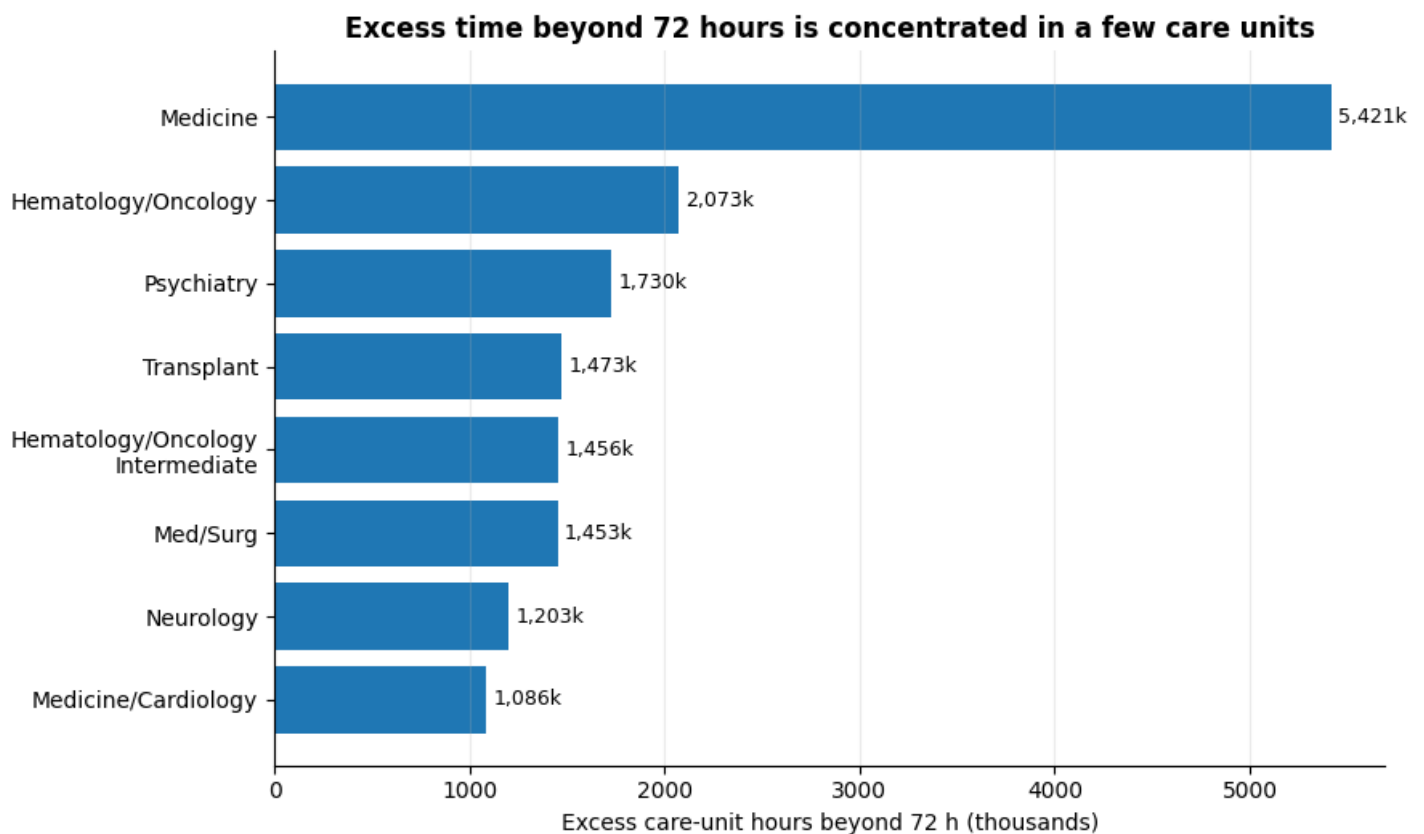


At the 72-hour threshold, **14.1%** of segments consume **65.3%** of all recorded segment-hours. This indicates that the main capacity burden is concentrated in the right tail rather than distributed evenly across all segments.

2. The burden is concentrated in specific care units

The second graph ranks care units by total hours accumulated after the first 72 hours of each long segment. The first 72 hours are not counted as excess.

A high rank identifies where the long-duration burden is located. It does not by itself establish inefficiency or avoidable delay; case mix, clinical complexity, transfer practices, transport, payer processes, and downstream capacity may all contribute.

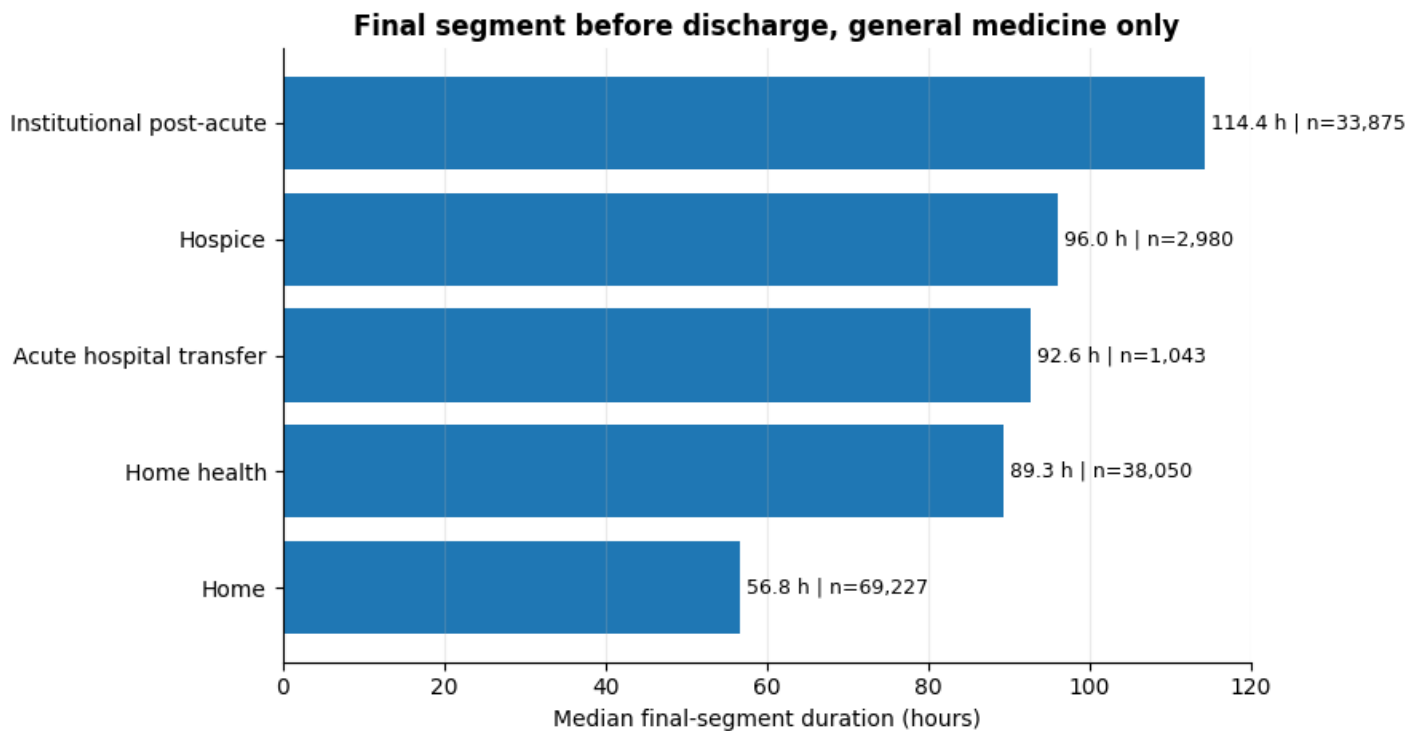


Medicine contributes approximately 5.4 million excess care-unit hours—more than twice the next-ranked unit—and is the first place an operations team should investigate.

3. Discharge destination provides a downstream-care signal

The comparison is limited to admissions whose last recorded service is general medicine (**MED**). It uses the direct median of final segments within each destination category.

Supporting signal: within admissions discharged from general medicine, the median final segment is **57.6 hours longer** for institutional post-acute destinations than for home discharge.



Bottom line

The operational problem is not the typical segment; it is the **small, resource-intensive right tail**.

The first place to investigate is **Medicine**, because it contributes the most excess time beyond 72 hours in the complete system-wide transfer cohort.

The median final segment in Medicine is **57.6 hours longer** for institutional post-acute destinations than for home discharge.

Next data needed: discharge-ready time, referral time, payer authorization, placement acceptance, downstream bed availability, transport booking, and actual departure. MIMIC can locate burden, but it cannot determine which hours were avoidable.

▼ Method, cohort, privacy, and interpretation notes

- System-wide scope: 1,867,366 valid segments. Of these, 1,458,484 are linked to 545,994 hospital admissions, while 408,882 have no `hadm_id`. Less than 11 segments over 365 days were excluded as probable administrative artifacts.
- **Headline/system-wide cohort:** every positive-duration row in `hosp.transfers` with non-null `careunit`, `intime`, and `outtime`, and duration no longer than 365 days. `hadm_id` may be null.
- **Admission-linked cohort:** the same cleaned transfer rows restricted to `hadm_id IS NOT NULL`. This cohort is used only in the downstream-care calculations.
- ED-only transfer rows without a hospital admission ID remain in the main hospital-wide distribution. Excluding them changes the median and creates the earlier 8.8-hour versus 16.6-hour discrepancy.
- Durations above 365 days are excluded as probable administrative artifacts.
- Quantiles are calculated with BigQuery `APPROX_QUANTILES`.
- “Excess time beyond 72 hours” is `max(duration - 72, 0)` summed across segments.
- The discharge-destination comparison uses the **direct median of final segments** among admissions whose last recorded service is `MED`; it does not average subgroup medians.
- Home health, hospice, acute-hospital transfer, and institutional post-acute care remain separate.
- Every grouped output requires `n ≥ 11`; no patient-level or row-level derived datasets are produced or shared. Only aggregate outputs with `n ≥ 11` are published.
- This is a single-center observational EHR analysis. Segment duration is not the same as confirmed discharge delay, medical readiness, avoidability, or causal impact.

Data source and citations

This report uses **MIMIC-IV v3.1**, a deidentified electronic health record dataset from Beth Israel Deaconess Medical Center covering patients admitted through the emergency department or an intensive care unit. The `hosp` module contains patient tracking data, including admissions and intra-hospital transfers.

- Johnson, A., Bulgarelli, L., Pollard, T., Gow, B., Moody, B., Horng, S., Celi, L. A., & Mark, R. (2024). *MIMIC-IV (version 3.1)*. PhysioNet. <https://doi.org/10.13026/kpb9-mt58>
- Johnson, A. E. W., Bulgarelli, L., Shen, L., et al. (2023). *MIMIC-IV, a freely accessible electronic health record dataset*. Scientific Data, 10, 1. <https://doi.org/10.1038/s41597-022-01899-x>
- Pollard, T., Moody, B. E., Lehman, L., et al. (2026). PhysioNet as a global platform for biomedical research. Nature Health. <https://doi.org/10.1038/s44360-026-00096-z>